# Reflection on Group 24 Assignment 2IRR00

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## 1 Project Description and Scope

For our project we want to make an audio book player that is accessible to elderly people. We want it to be able to play audio books that are imported into the program. While playing the audio book, it can be paused and started, jump to an a chosen section and have the volume be adjusted. It should also have an overview of all series available and a view within a series. Currently the project is functional with most of our planned features present. It may be that not everything will be finished, regardless we have succeeded in making a functional program.

## 2 Project Reflection

### 2.1 Process and Collaboration

Our group has a lot of enthusiasm for the project and we all wanted to jump right in. However initially we did have some trouble with clear communication and organizing ourselves and we needed a moment to figure out how we want to approach the project. After that we were able to divide tasks and work on them independently. Near the end of the project where we were integrating the different parts of the project we realized that we were actually a bit more dependent on each other than expected. One of the issues was how we were handling the audio book objects internally. For example some of us needed to generate/load audio books for their respective parts, which led to repeated implementations of the same feature. Also how we passed around objects was different per person, which made the integration phase more difficult. In the end it did work out with some rewriting and refactoring, but it would have been better, if we decided before hand how to deal with them and merged that feature early on into everyone's branches.

### 2.2 Testing and Quality Assurance

Near the start of the project we agreed to work on the project through git branches. Before any branch could be merged with the main brach, someone else would review the branch. This included reading the code, running the program and looking at the unit tests. This ensured when there were issues with the branches, it was found on time and fixed before being merged with the main branch.

### 2.3 Design Principles and Patterns

For our project we want to have settings and a program state that are saved and restored when the program is closed and opened again. For this we used the singleton design pattern, as we want to have only one settings object through out the entire program. The example in the lecture gave us good solution to the problem [1]. Next we also used the observer listener design pattern. However this was less a choice and more a requirement for the UI. We are using JavaFX for the user interface and some elements require the use of observable objects. To my knowledge we have not applied any of the other design patterns presented in the lectures. However we did try to keep the project as simple as it needed to be to reach our goals.

## 3 Self-Reflection

I think my biggest contribution to the project are my organizational skills. This helped others to contribute more effectively. During the project I was mostly able to keep the group on the same page and have the meetings be clear and concise. During the project there was a private issue going in the background, so on the software development I was not able to contribute as much as I wanted to. I was able to clearly communicate this to the group, so the others could step in on time, which allowed the project to keep moving on schedule.

## 4 Conclusion and Learnings

In summary I learned I can organize a group of people quite well. Specifically for software design I have learned to be more aware of where and how internal dependencies can cause problems. I have learned to apply particular design patterns en effectively solve programming problems.

## References

[1] Jacob Krüger. 2025. Software Design Course. (2025). Lecture: Building on Established Solutions to Create Objects.